

SE 216 – SOFTWARE PROJECT MANAGEMENT
SOFTWARE TOOLS DOCUMENT

PROJECT NAME: AI Personal Health & Fitness Coach

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TASK #	PROJECT TASKS WHICH REQUIRE SOFTWARE TOOL SUPPORT
1	Development of the AI Personalized Recommendation & Optimization Engine
2	Scalable Backend Infrastructure & Health Data Security Management
3	Development Environment Setup for AI and Backend Implementation

SOFTWARE TOOLS FOR TASK 1:Development of the AI Personalized Recommendation & Optimization Engine

Tool Cost/Training/Functionality Data

Tool	Hugging Face	vLLM	AWS SageMaker	Azure AI Studio
Cost	\$0 (Open Source)	\$0 (Open Source)	\$500/mo	\$420/mo
Training Days	5	3	12	10
Functionality	10	6	8	9

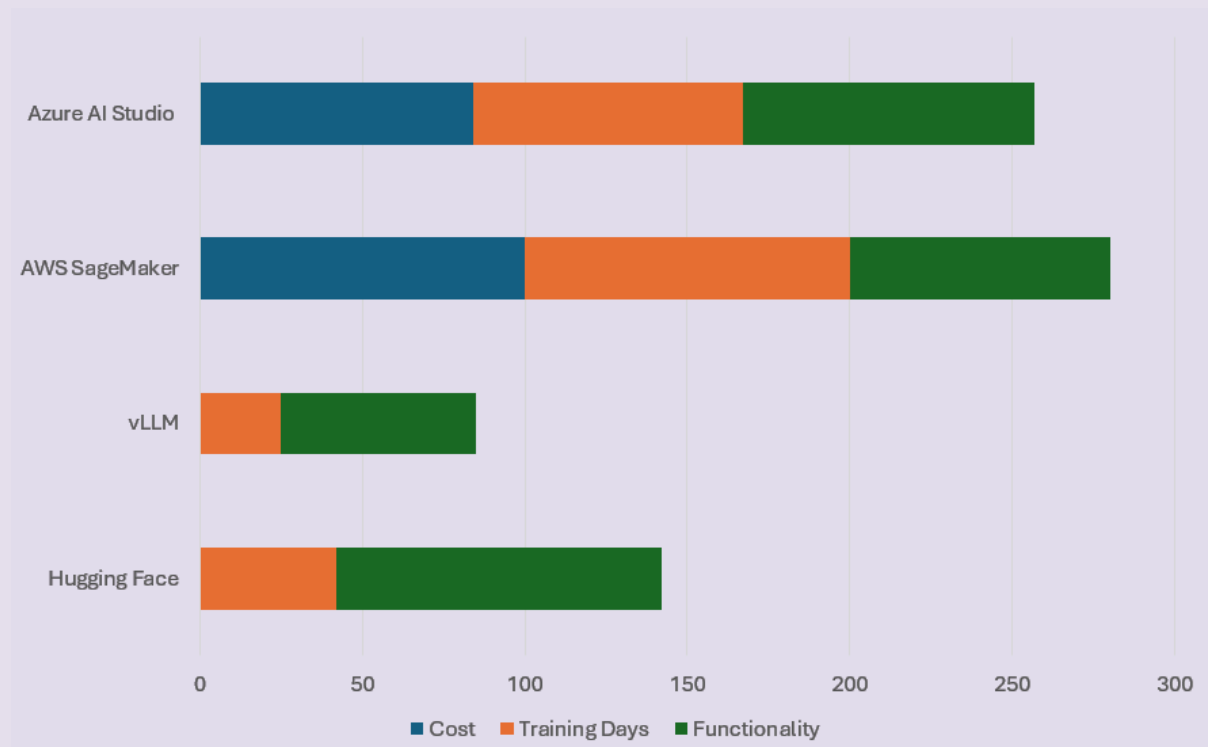
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Normalized Cost/Training/Functionality Data

Tool	Hugging Face	vLLM	AWS SageMaker	Azure AI Studio
Cost	0	0	100	84
Training Days	42	25	100	83
Functionality	100	60	80	90

Normalized Tool Graph



Which tool has been selected? Why?

Hugging Face was selected.

Why: Getting pre-trained models up and running quickly was our top priority here, alongside painless fine-tuning. Hugging Face ended up being the obvious pick. Rather than spending almost two weeks wrestling with massive cloud environments like SageMaker just to deploy a tweaked model, their ecosystem lets us pull pre-trained weights, fine-tune them on our custom dataset, and get everything live in a fraction of the time. Plus, keeping the

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base cost at zero while still getting top-tier usability gives the team way more breathing room when we inevitably need to scale up the infrastructure later.

SOFTWARE TOOLS FOR TASK 2: Scalable Backend Infrastructure & Health Data Security Management

Tool Cost/Training/Functionality Data

Tool	Firebase	PostgreSQL	MongoDB Atlas	Microsoft Azure SQL
Cost	20/mo	30/mo	100/mo	120/mo
Training Days	3	10	5	12
Functionality	8	9	8	10

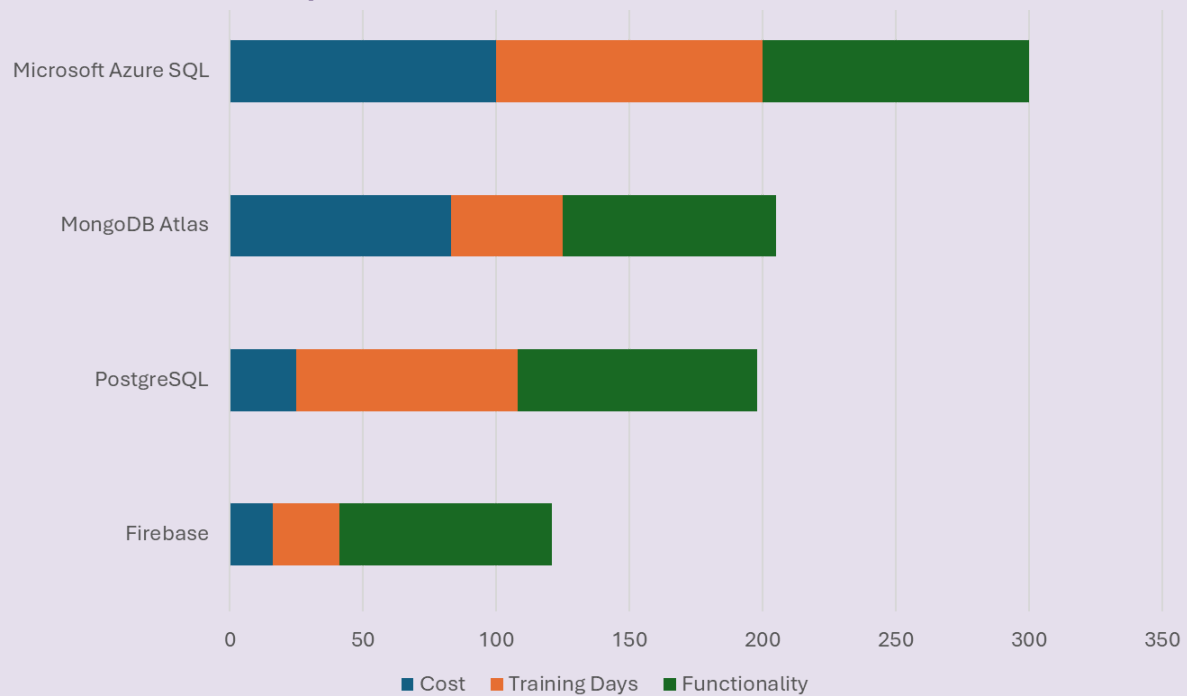
Normalized Cost/Training/Functionality Data

Tool	Firebase	PostgreSQL	MongoDB Atlas	Microsoft Azure SQL
Cost	16	25	83	100
Training Days	25	83	42	100
Functionality	80	90	80	100

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Normalized Tool Graph



Which tool has been selected? Why?

PostgreSQL was selected.

Why: PostgreSQL was selected for its strong ability to manage large-scale structured data, support complex queries, and ensure high data integrity. It guarantees that database operations are processed reliably and consistently, meaning that data is not lost, partially updated, or left in an inconsistent state even in case of system failures. This level of reliability is essential for handling sensitive health and fitness data. Additionally, PostgreSQL provides advanced features such as joins, indexing, and high scalability, making it a robust and efficient solution for backend development.

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SOFTWARE TOOLS FOR TASK 3:Development Environment Setup for AI and Backend Implementation

Tool Cost/Training/Functionality Data

Tool	PyCharm Professional	Wing IDE Pro	Eclipse	Visual Studio
Cost	\$1500/yearly	\$690	\$0	\$3240/yearly
Training Days	6	5	8	4
Functionality	10	8	7	8

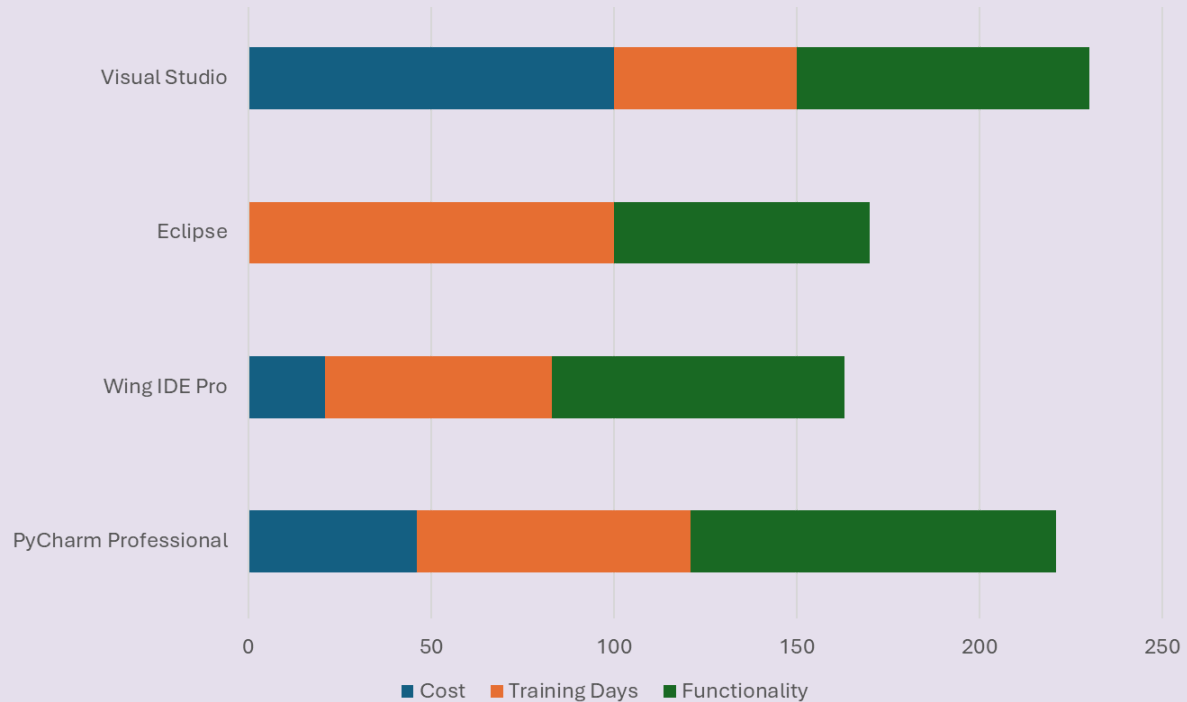
Normalized Cost/Training/Functionality Data

Tool	PyCharm Professional	Wing IDE Pro	Eclipse	Visual Studio
Cost	46	21	0	100
Training Days	75	62	100	50
Functionality	100	80	70	80

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Normalized Tool Graph



Which tool has been selected? Why?

PyCharm Professional was selected as the development environment for Python because it provides advanced features such as intelligent code completion, powerful debugging tools, integrated testing support, database management, and seamless integration with AI and backend frameworks. Although it is a paid solution, its high functionality and enterprise-level capabilities make it the most suitable IDE for large-scale and AI-based projects.